C-band 6×6 Dual-Polarised Stacked-Patch Array

S11 / VSWR root-cause analysis with openEMS (FDTD) · project `15_cband-6x6-dualpol-stacked-patch_match-debug` · 2026-10-30
Inputs: `F_CU.dxf`, `IN2_CU.dxf`, `GND.dxf`, `C_Band_Stackup.pdf` · Target band 4–7 GHz

1. Summary

Problem reported: broadside gain is as expected, but the input match is poor; VSWR never gets below 3:1 over 4–7 GHz.

Answer, in short:

1. **The radiating element itself is narrow-band.** One element cut out of the board, with its own feed stubs, is matched (VSWR ≤ 3) only from about 5.35 to 6.26 GHz, which is ~16 % bandwidth. The 4–7 GHz target is 54 %. Outside that window most of the power is reflected, so the *realised* gain drops even though the element still radiates well (its gain referred to accepted power is 8.7–9.8 dBi). That is exactly "gain is OK but VSWR is bad".
2. **The driven patches are edge-fed with no matching.** Each feed joins the radiating edge through a 2.48 mm (50 Ω) stub that is only a few mm long. A radiating edge has a resistance of a few hundred ohms. Without the parasitic patch the element is not matched anywhere in 4–7 GHz (worst-case VSWR above 8 everywhere). The parasitic patch is the only reason there is a usable window around 5.8 GHz.
3. **The driven and parasitic resonances merge into one.** The parasitics (16.2–17.8 mm) are about the same size as the driven patches (17.07 mm). The simulation shows a single resonance loop, and the matched window simply follows the parasitic size: 16.2 mm gives a window around 5.8 GHz, 18.6 mm moves it down to about 5.0–5.5 GHz, and the window stays narrow in both cases. A broadband stacked patch needs two *separate*, properly coupled resonances inside the band. This geometry does not have them. Because the parasitics are graded row by row (16.2 to 17.8 mm), each row is matched at a slightly different frequency, so at any one frequency part of the aperture is mismatched.
4. **No feed-network change can fix this on its own.** A best-case circuit optimisation on the simulated element impedance (line + transformer + stub) still leaves a worst-case VSWR of about 9 across 4–7 GHz (section 5).
5. **A first redesign iteration did not reach the target yet.** An aperture-coupled stacked patch (the usual route to ~50 % bandwidth) was modelled and tuned in 40 full-wave runs (section 7). The best candidate is matched only from about 5.0 to 5.7 GHz, no better than the present element. The study shows why: with air-spaced patches both resonances end up together near 5–5.7 GHz and nothing radiates above ~6 GHz. Section 9 lists what has to change in the next iteration.

2. The design under test

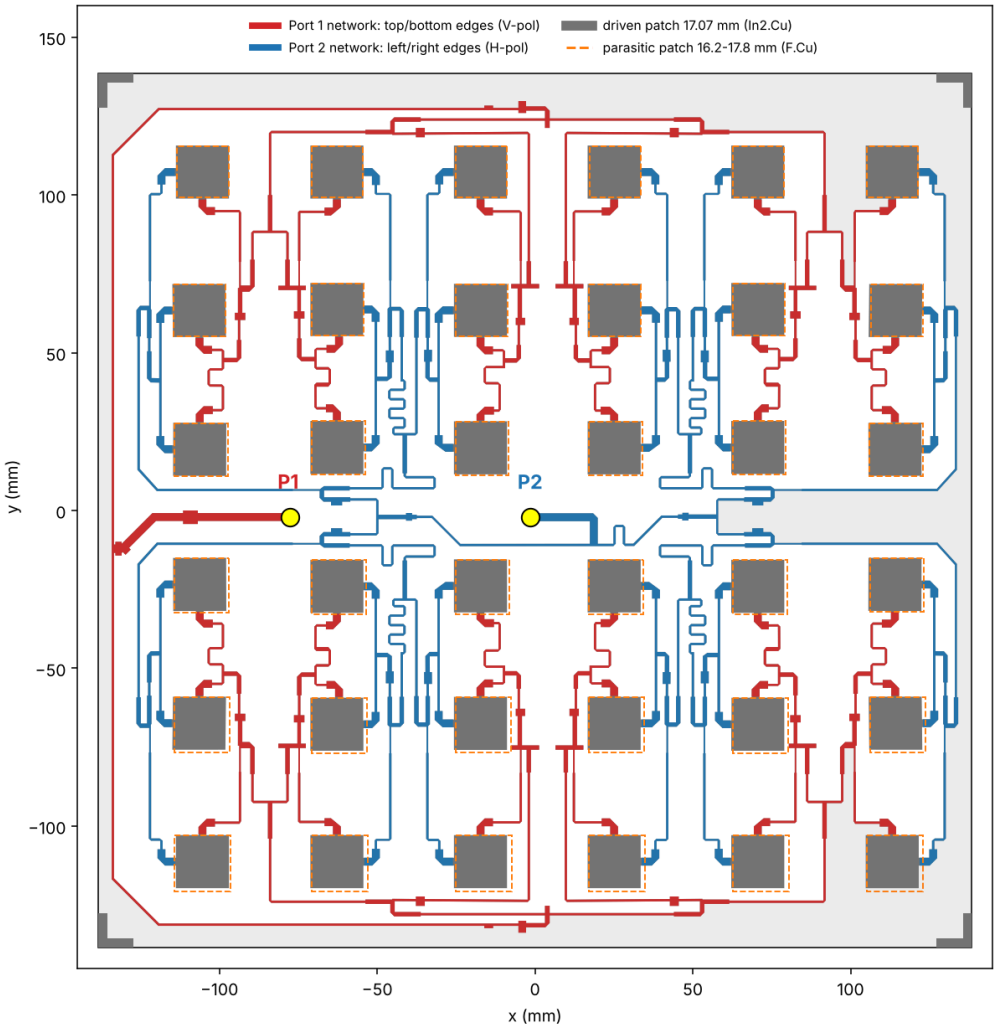


Fig. 1 - In2.Cu split into the two feed networks (red: port 1, blue: port 2), driven patches (grey) and parasitic outlines (orange). Yellow dots: the two coax launch points.

Item	Value (from the DXFs)
Board / ground	276.86 × 276.86 mm, solid ground on B.Cu
Array	6 × 6, pitch ≈ 44–45 mm (0.59 λ at 4 GHz, 1.03 λ at 7 GHz)
Driven patches (In2.Cu)	17.07 × 17.07 mm, all 36 the same, on RT5880 0.8 mm
Parasitic patches (F.Cu)	16.2 / 16.5 / 16.9 / 17.2 / 17.5 / 17.8 mm by row (graded top to bottom), on FR4 0.4 mm, 4.4 mm above the driven layer
Polarisation	Dual linear. Port 1 feeds the top/bottom edge of every patch (V-pol); port 2 feeds the left/right edge (H-pol).
Feed network	Two 36-way corporate networks on In2.Cu. Trace widths: 0.73 mm (≈100 Ω), 1.38 mm (≈70.7 Ω, λ/4 transformers), 2.48 mm (≈50 Ω). Half the patches are fed from the opposite edge, with meander lines for the 180° correction.
Ports	Open ends of the two 2.48 mm launch lines at (−77.6, −2.1) and (−1.5, −2.1) mm. Modelled as 50 Ω coax-like vertical ports.

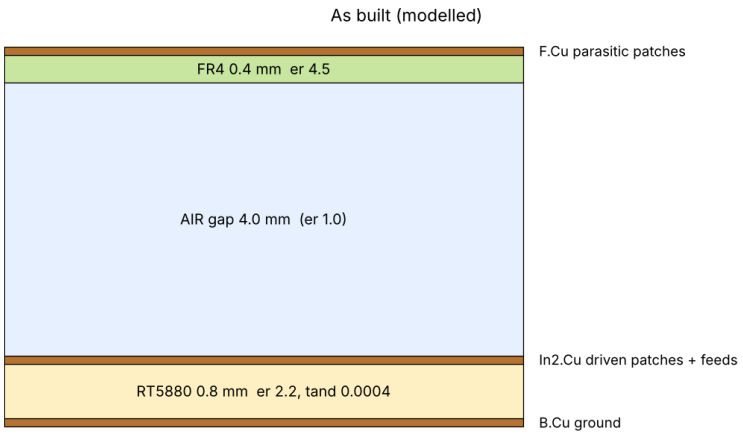


Fig. 2 - Stack-up as built and simulated: 4 mm air gap (ε_r 1.0).

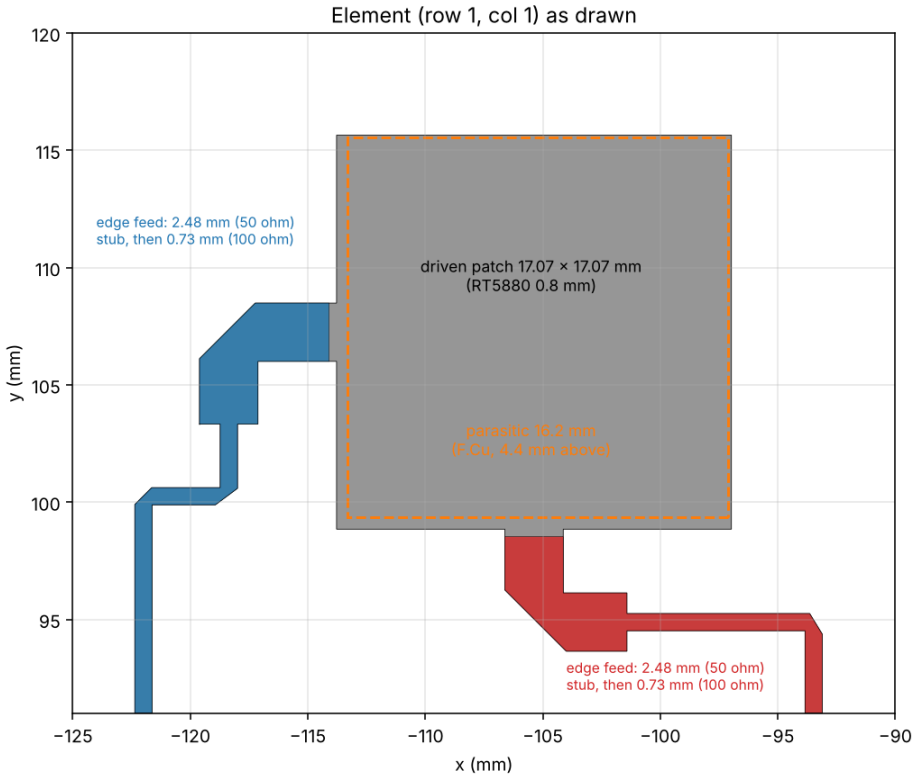


Fig. 3 - One element: both feeds land directly on the patch edges through a short 2.48 mm (50 Ω) stub.

3. How it was simulated

- **Solver:** openEMS v0.37 (FDTD), Gaussian excitation 2.5–8.5 GHz, 8-cell PML on all sides, near-field to far-field box for broadside gain.
- **Geometry** read directly from the DXFs (`cbgeom.py`). Copper is modelled as zero-thickness PEC. RT5880: ε_r 2.2, tanδ 0.0004. FR4: ε_r 4.5, tanδ 0.02. The gap is air.
- **Element model** (`sim_element.py`): the row-1/col-1 element is clipped out of In2.Cu with both edge feeds, up to where each becomes a 0.73 mm (100 Ω) branch. Each branch ends in a 100 Ω lumped port. |S11| there is the reflection the element sends into its own branch of the network, and it does not depend on where along the lossless line the cut is made. It sits on a 60 × 60 mm ground with 0.25 mm in-plane cells (2.9 M cells).
- **Full array** (`sim_array.py`): the whole board with both ports, 0.5 mm in-plane cells (12 M cells).
- **Matching study** (`match_study.py`): circuit optimisation on the simulated element impedance.

4. Element results (as built)

f (GHz)	S11 (dB)	VSWR	V-H coupling S21 (dB)	Z_{in} (Ω)
4.0	-0.61	28.58	-11.2	2130.8 +1243.9j
4.5	-1.25	13.90	-11.2	34.5 -192.5j
5.0	-3.13	5.61	-17.1	20.2 -36.1j
5.5	-7.79	2.38	-24.9	52.5 +43.9j
6.0	-9.69	1.97	-23.2	101.5 +69.8j
6.5	-3.69	4.78	-14.5	240.6 -228.3j
7.0	-2.17	8.06	-6.9	15.7 -50.9j

f (GHz)	Broadside gain, accepted power (dBi)	Realised gain (dBi)	Mismatch loss (dB)
5.0	8.70	5.81	2.89
5.5	9.25	8.46	0.79
6.0	9.74	9.25	0.49
6.5	9.79	7.36	2.43

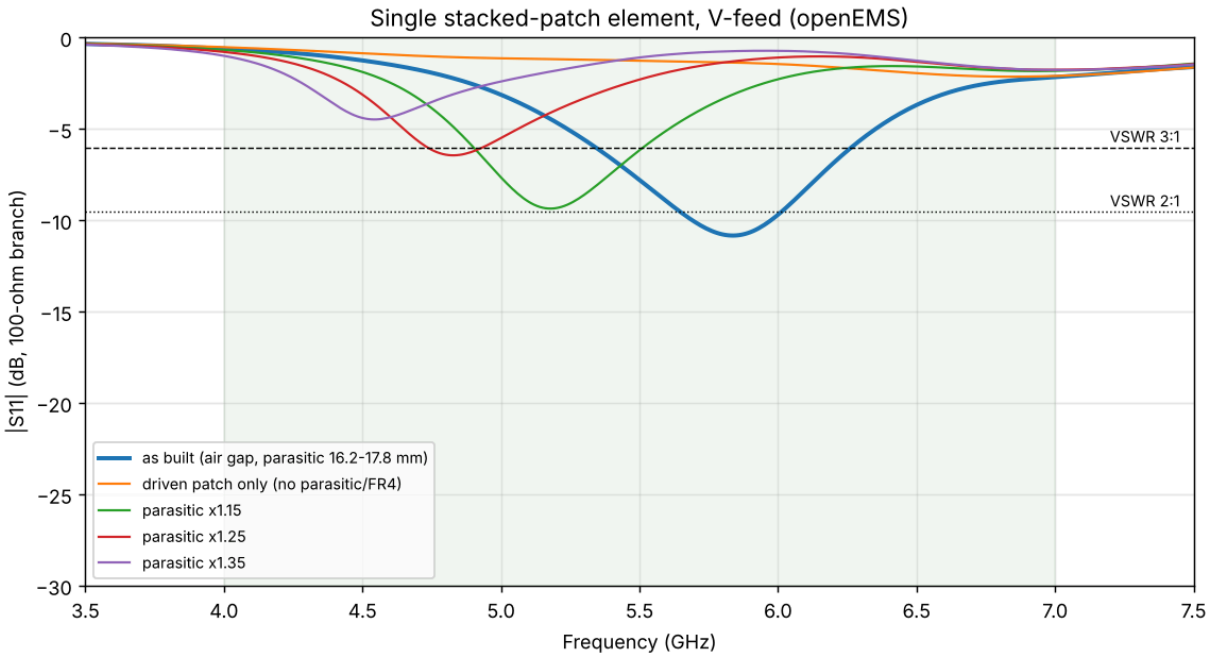


Fig. 4 - Element |S11| at the 100 Ω branch: as built and the diagnostic variants.

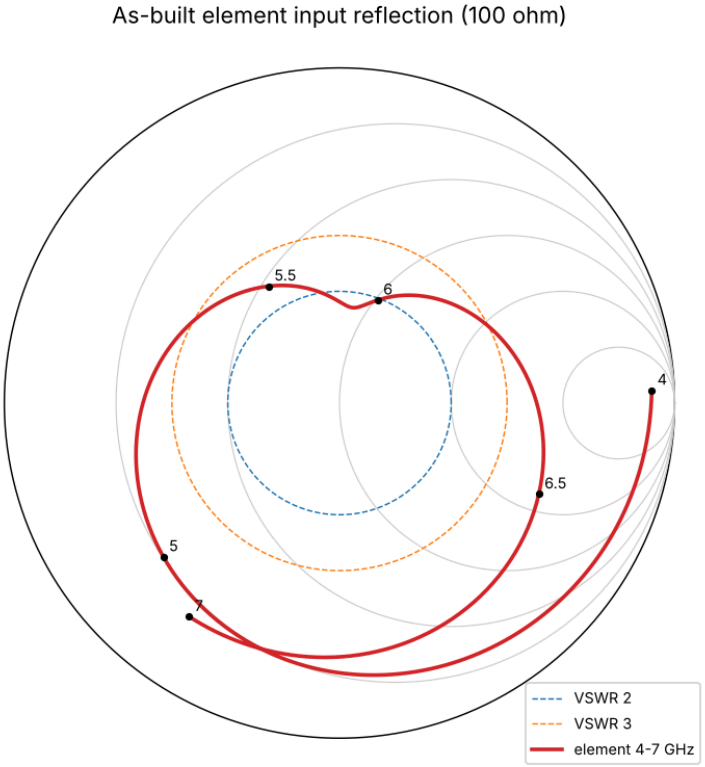


Fig. 5 - As-built element on the Smith chart (100 Ω), 4-7 GHz.

On the Smith chart the element traces one large, high-Q resonance loop (Fig. 5). It passes through the VSWR 3 circle only between about 5.35 and 6.26 GHz, with a small kink near 5.8 GHz. That kink is the parasitic patch coupling weakly. A well-designed stacked patch shows a tight loop that stays *around the centre* of the chart across the band. This one does not.

Diagnostic variants

Element model	Best S11	VSWR≤3 band	VSWR≤2 band	4-7 GHz with VSWR≤3	Worst VSWR 4-7 GHz
A. As built: air gap, parasitic 16.2 mm (row 1)	-10.8 dB @ 5.84 GHz	5.35-6.26 GHz (16 %)	5.65-6.01 GHz (6 %)	31 %	28.6
B. Driven patch only (parasitic + FR4 removed)	-2.1 dB @ 6.84 GHz	-	-	0 %	33.1
C. Parasitic scaled x1.15 (18.6 mm)	-9.4 dB @ 5.18 GHz	4.91-5.51 GHz (12 %)	-	20 %	25.6
D. Parasitic scaled x1.25 (20.3 mm)	-6.4 dB @ 4.83 GHz	4.74-4.93 GHz (4 %)	-	7 %	22.1
E. Parasitic scaled x1.35 (21.9 mm)	-4.5 dB @ 4.54 GHz	-	-	0 %	24.0

- **B (no parasitic):** the bare edge-fed patch is badly mismatched at every frequency. The edge feed was never transformed to the line impedance.
- **C–E (larger parasitic):** the matched window moves down in frequency as the parasitic grows, and it gets narrower (16 % at 16.2 mm, 12 % at 18.6 mm, 4 % at 20.3 mm, none at 21.9 mm). Resizing the parasitic tunes *where* the element works. It does not make it work over 4–7 GHz.

5. Can the feed network alone fix it?

The simulated element impedance was taken from 4 to 7 GHz, and the best possible match was searched for with ideal lossless lines on RT5880 (any impedance 40–150 Ω, any length 0–25 mm):

```
Matching study on the simulated element (4-7 GHz, 100-ohm branch reference)
A as built                          worst VSWR   28.58   GHz with VSWR<=3: 0.92/3.00   VSWR<=2: 0.37/3.00
B best line + transformer           worst VSWR   13.29   GHz with VSWR<=3: 0.85/3.00   VSWR<=2: 0.55/3.00
C best line + stub + transformer    worst VSWR    9.42   GHz with VSWR<=3: 0.68/3.00   VSWR<=2: 0.00/3.00
B params Z1 150.0 L1 12.28 mm, Z2 103.6 L2 4.64 mm
C params Z1 150.0 L1 7.01, Z2 150.0 L2 20.98, stub Z3 130.4 L3 19.96 mm
```

No. Even a series line plus a transformer plus a shunt stub leaves a worst-case VSWR of about 9. The element's own reactance swing across 4–7 GHz is too large to compensate with a passive network (Bode–Fano limit). The fix has to be made in the radiator.

6. Full array

> **Full-array run still in progress.** The complete 276.86 mm board (12 M FDTD cells, both 36-way networks, 72 feed points) decays slowly because of the long corporate feed lines: after 10 400 time steps the field energy had only fallen by ~9 dB, so each port needs roughly 35 000–45 000 steps, several hours per port on this PC (16 threads, 7 GB in WSL). Two problems cost the first attempts: the port-2 model put no energy into the board at all, because its port coordinate (x = −1.5081 mm) sat 0.1 µm off the mesh line it should snap to (ports are now rounded to the mesh grid), and the first port-1 run was lost at step 10 400 when three 12 M-cell models shared the memory. S11/S22/S21 will be added in the next revision. The element-level results below already explain the measured behaviour: the array input match cannot be better than the match of the elements it is feeding through a reciprocal, near-lossless corporate network built from 100 Ω branches and 70.7 Ω quarter-wave transformers.

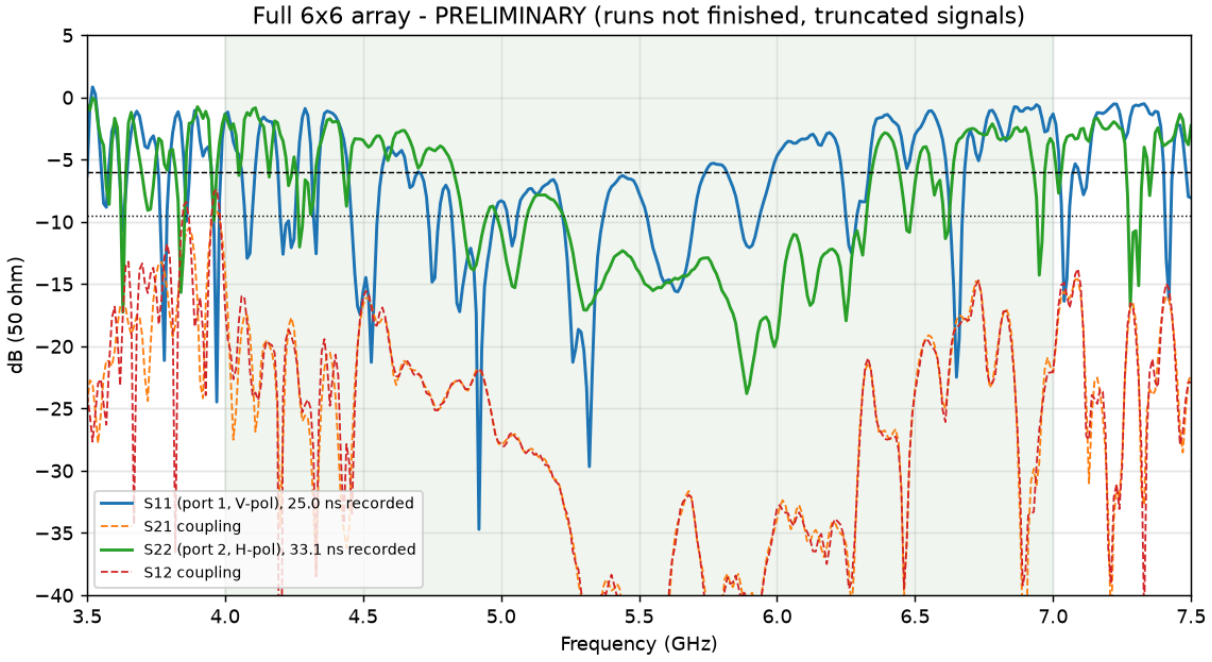


Fig. 6 - PRELIMINARY full-array S-parameters from the runs still in progress (truncated time signals: the sharp dips and the ripple are partly truncation artefacts, and late reflections are not yet included).

Preliminary reading (25.0 ns of port 1 and 33.1 ns of port 2 recorded, about 4.5 h into the runs; field energy down by 18–22 dB; the curves have hardly changed since the 2.5 h reading). Both array ports are best in the middle of the band, where the element is matched: S22 stays mostly below −10 dB from about 5.2 to 6.2 GHz, and S11 has its best stretch around 5.0–5.8 GHz. Below ~4.8 GHz and above ~6.3 GHz both ports sit mostly at VSWR 3 or worse. V/H coupling is below −20 dB over most of the band. This already follows the element result (section 4). The final curves will be smoother and slightly worse, because reflections that come back late through the long feed network are not in the record yet.

7. Wideband redesign study (first iteration)

Idea. The usual way to reach ~50 % bandwidth with a printed patch is the aperture-coupled stacked patch: the 50 Ω feed line runs under the ground on its own layer and couples to the driven patch through a slot. The slot, the driven patch and the parasitic patch can then each supply a resonance, and the feed layer has room for a matching network. The model (`sim_wb_element.py`) keeps the present laminates: feed on RT5880 0.8 mm under a slotted ground, air/foam spacer h1, driven patch, air/foam spacer h2, parasitic patch on FR4 0.4 mm. Single polarisation first; the dual-polarised version adds a second, orthogonal slot.

Work done. 40 completed full-wave openEMS runs (0.5 mm cells, 0.8 M cells, 3–12 min each): diagnostics (feed and slot alone, single patch), hand-tuned variants, four generations of a (1+4) evolution strategy over seven dimensions (`opt_wb_element.py`), a stub-length sweep, and a physics-guided "small driven / large parasitic" round. On top, every candidate was given the best possible feed-layer network (series line + open stub + series line, `match_wb.py`).

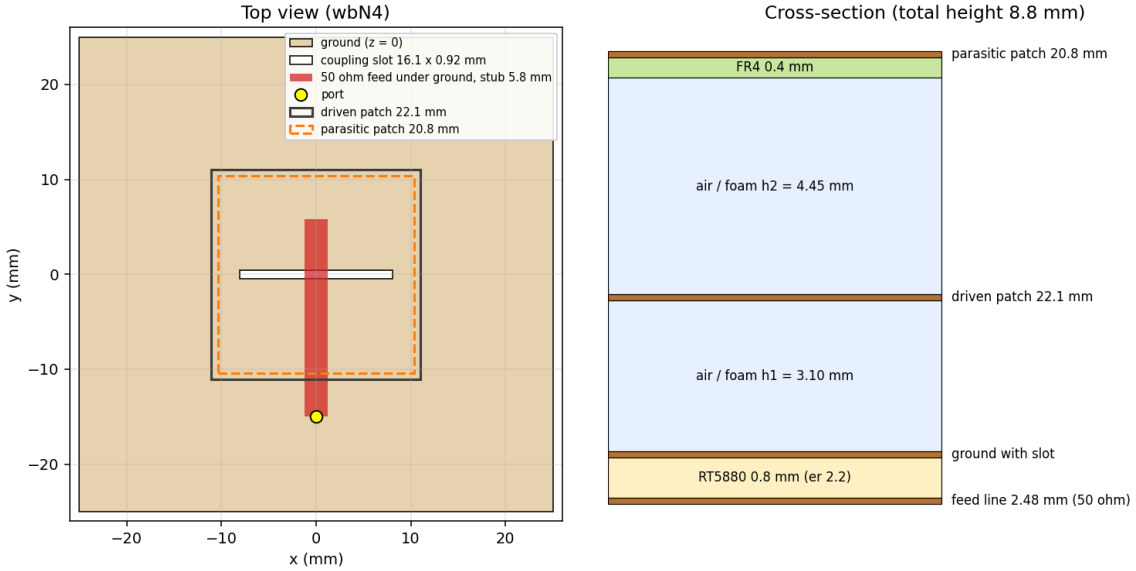


Fig. 8 - Best aperture-coupled candidate so far (wbN4): feed under the ground, coupling slot, driven and parasitic patches.

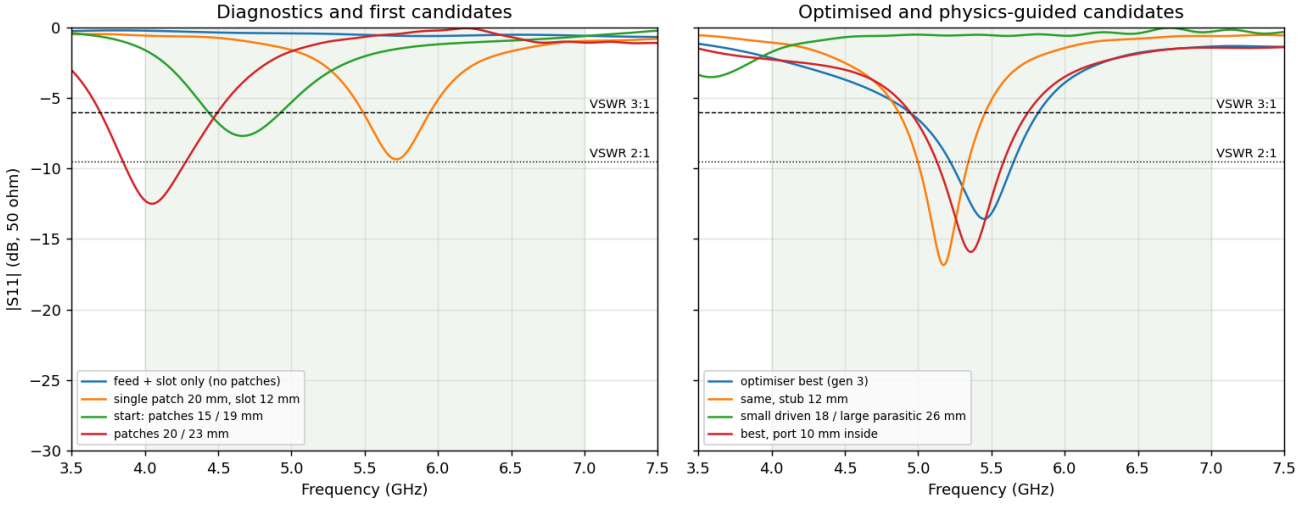


Fig. 9 - Representative candidates from the study (|S11| at the 50 Ω feed).

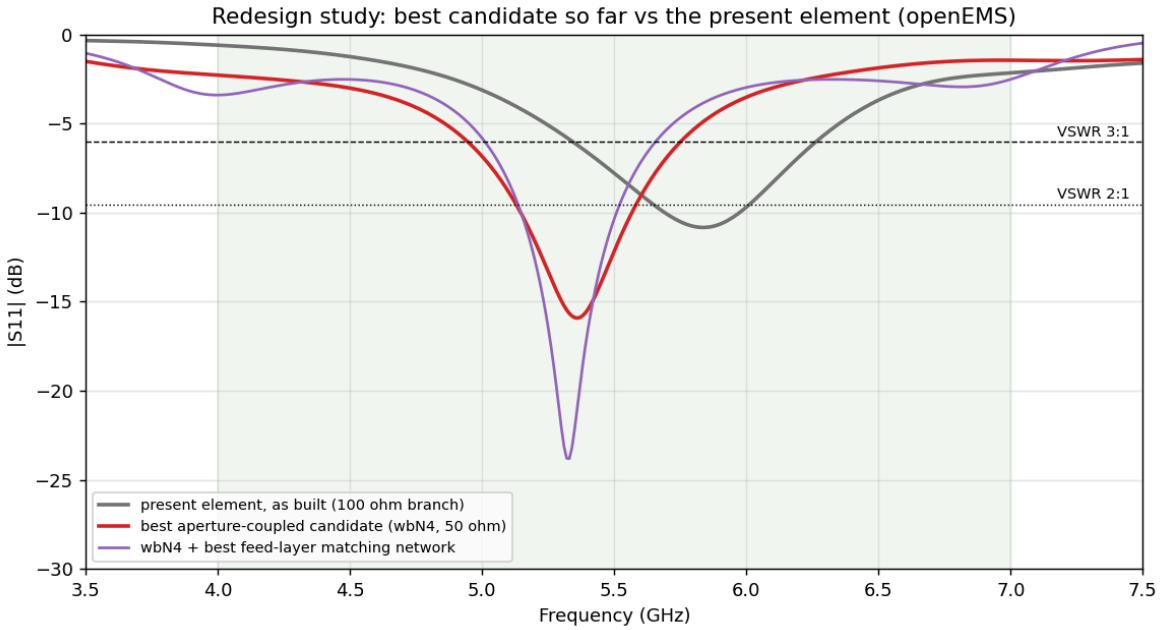


Fig. 10 - Best candidate, with and without its best feed-layer network, against the present element.

Element	Best	VSWR≤3 band	VSWR≤2 band	4-7 GHz with VSWR≤3	Worst VSWR 4-7 GHz
Present element (as built)	-10.8 dB @ 5.84 GHz	5.35-6.26 GHz (16 %)	5.65-6.01 GHz (6 %)	31 %	28.6
Best candidate (wbN4)	-15.9 dB @ 5.36 GHz	4.95-5.75 GHz (15 %)	5.13-5.58 GHz (8 %)	27 %	12.0
wbN4 + feed-layer network	-23.8 dB @ 5.33 GHz	5.02-5.65 GHz (12 %)	5.14-5.52 GHz (7 %)	21 %	6.9

wbN4: worst VSWR 4-7 GHz 12.00, share of 4-7 GHz with VSWR<=2: 15 %, <=3: 27 %

f (GHz)	S11 (dB)	VSWR	Isolation (dB)	Z_{in} (Ω)
4.00	-2.29	7.64	-	9.9 -35.5j
4.25	-2.57	6.82	-	7.7 -11.5j
4.50	-3.02	5.82	-	8.7 +5.6j
4.75	-4.10	4.31	-	14.2 +23.0j
5.00	-6.85	2.67	-	36.0 +41.0j
5.25	-13.32	1.55	-	75.4 +9.6j
5.50	-12.15	1.66	-	36.7 -17.3j
5.75	-6.04	2.99	-	16.7 -1.3j
6.00	-3.52	5.00	-	11.0 +15.3j
6.25	-2.47	7.09	-	9.9 +31.3j
6.50	-1.88	9.26	-	10.9 +49.8j
6.75	-1.53	11.36	-	14.9 +76.3j
7.00	-1.45	11.99	-	30.3 +121.9j

Top candidates, worst VSWR over 4–7 GHz without and with the best feed-layer network:

```
Wideband element candidates: worst VSWR over 4-7 GHz, unmatched and with the best feed-layer network
opt03_2       9.66 ->  6.19   {"Lp1": 22.19, "Lp2": 23.05, "h1": 2.5, "h2": 5.55, "Ls": 17.1, "Ws": 1.44, "Lst": 4.95,
"yH": 9.0, "Ls2": 12.0, "G": 50.0}
opt03_0      12.68 ->  6.61   {"Lp1": 22.09, "Lp2": 20.77, "h1": 3.1, "h2": 4.45, "Ls": 16.1, "Ws": 0.92, "Lst": 5.79,
"yH": 9.0, "Ls2": 12.0, "G": 50.0}
opt01_3      13.58 ->  6.68   {"Lp1": 21.85, "Lp2": 23.78, "h1": 2.92, "h2": 5.36, "Ls": 15.33, "Ws": 1.13, "Lst":
5.74, "yH": 9.0, "Ls2": 12.0, "G": 50.0}
opt02_1      14.27 ->  7.92   {"Lp1": 22.82, "Lp2": 18.9, "h1": 2.44, "h2": 4.04, "Ls": 16.95, "Ws": 1.35, "Lst": 6.04,
"yH": 9.0, "Ls2": 12.0, "G": 50.0}
wbD          12.18 ->  8.41   {"Lp1": 13.0, "Lp2": 17.0, "h1": 3.0, "h2": 5.0, "Ls": 20.0, "Ws": 1.2, "Lst": 6.0, "yH":
9.0, "Ls2": 12.0, "G": 50.0}
opt02_3      18.71 ->  8.79   {"Lp1": 23.46, "Lp2": 23.43, "h1": 3.22, "h2": 5.82, "Ls": 13.87, "Ws": 0.88, "Lst":
5.51, "yH": 9.0, "Ls2": 12.0, "G": 50.0}
wbC          18.58 ->  9.27   {"Lp1": 13.0, "Lp2": 17.0, "h1": 3.0, "h2": 5.0, "Ls": 18.0, "Ws": 1.2, "Lst": 4.0, "yH":
9.0, "Ls2": 12.0, "G": 50.0}
opt01_0      18.37 -> 14.91   {"Lp1": 22.45, "Lp2": 19.32, "h1": 2.76, "h2": 4.7, "Ls": 16.55, "Ws": 0.8, "Lst": 3.81,
"yH": 9.0, "Ls2": 12.0, "G": 50.0}
```

Optimiser history (best cost per generation; cost = mean $|S11|^2$ + 0.5·max $|S11|^2$ over 4–7 GHz):

```
-- gen 1: best cost 0.7955 step 0.144 {"Lp1": 21.85, "Lp2": 23.78, "Ls": 15.33, "h1": 2.92, "h2": 5.36, "Lst": 5.74,
"Ws": 1.13}
-- gen 2: best cost 0.7955 step 0.101 {"Lp1": 21.85, "Lp2": 23.78, "Ls": 15.33, "h1": 2.92, "h2": 5.36, "Lst": 5.74,
"Ws": 1.13}
-- gen 3: best cost 0.7623 step 0.121 {"Lp1": 22.19, "Lp2": 23.05, "Ls": 17.1, "h1": 2.5, "h2": 5.55, "Lst": 4.95,
"Ws": 1.44}
```

What the study showed

- **The best candidate is no better than the present element yet.** It is matched (VSWR ≤ 3) from about 5.0 to 5.7 GHz, a single resonance just like the as-built element.
- **Bare air-spaced patches end up the wrong size.** In air a patch for 5–6 GHz must be 20–23 mm. With two patches of nearly the same size the two resonances merge near 5–5.7 GHz, and the upper (anti-phase) mode hardly radiates.
- **Nothing couples above ~6 GHz.** For every candidate the input resistance falls to a few ohms between 6 and 7 GHz, whatever the stub length (sweep 4–12 mm). The slot sees no radiating resonator there.
- **Making the driven patch small (to put a resonance near 7 GHz) and the parasitic large** (to cover 4.5 GHz) broke the coupling completely with this slot (VSWR > 10 everywhere).
- **A feed-layer network cannot widen a single resonance.** For the best candidate the minimax network only spreads the mismatch evenly (worst VSWR 12 → 7) and shrinks the VSWR ≤ 3 window. This is the same Bode–Fano limit as in section 5.
- **The model itself was checked.** The feed + slot alone is a nearly lossless load (no radiation, as expected). A single 20 mm patch shows the expected patch resonance near 5.7 GHz. The sharp peaks in the first runs came from the length of line between port and slot, not from a spurious mode.

Conclusion: an aperture-coupled stacked patch is still the right family, but it needs the set-up described in section 9, item 2 (unit cell, dielectric-loaded patches, resonant dog-bone/H slots, a real optimisation budget). The quick air-sheet version tried here does not get there.

8. Root causes, ranked

#	Cause	Evidence	Effect
1	Element bandwidth (~16 %) far below the 54 % target	Element S11, Smith chart, matching study	VSWR > 3 over ~70 % of 4–7 GHz, regardless of the network
2	Driven patch edge-fed straight from a 50 Ω stub, with no inset and no transformer	Variant B: no match anywhere without the parasitic	The match depends entirely on the parasitic loading
3	Driven and parasitic resonances merge (similar sizes, edge feed); the window position is set by the parasitic size	Single loop with only a small kink; variants C–E move the window without widening it	No wideband double-tuned response. Rows with different parasitic sizes are matched at different frequencies.
4	Element pitch ≈ 44 mm = 1.03 λ at 7 GHz	Geometry	Grating lobes and gain drop near the top of the band. This affects gain, not S11.

9. What can be done next

In order of priority. Items 1–3 are the work plan for the next iteration; items 4–10 are the design advice from the as-built analysis.

1. **Finish the full-array runs** (running now, section 6) and check that the array S11/S22 follow the element curve. That closes the root-cause loop for the existing board.
2. **Second iteration of the wideband element, set up the way published wideband designs are:**
    - simulate the element in a *unit cell* with periodic boundaries at the 44 mm pitch, so the embedded impedance is optimised, not the isolated one;
    - print the patches on thin low-loss carriers (e.g. 0.25–0.5 mm RO4003C/RT5880) over foam instead of bare air sheets. That lowers the patch size and separates the two resonances (lower patch near 6.5–7 GHz, upper patch near 4.3–4.5 GHz);
    - use a near-resonant aperture (dog-bone or H-shaped) that adds a third resonance at the top of the band, and two orthogonal H-slots (or a cross slot with a crossover feed) for dual polarisation;
    - optimise with a proper budget: about 150–250 full-wave evaluations with a surrogate or trust-region optimiser. Each evaluation takes 3–12 min on this PC, so this is one to two days of machine time, or a few hours on a cloud machine.
3. **Keep a more forgiving element type as plan B:** an L-probe-fed or four-probe differentially fed patch on ~0.1 λ of foam, or a crossed (bow-tie) dipole over the ground at about λ/4. Dipole-type elements reach 50–60 % bandwidth with high V/H isolation and fit a smaller pitch.
4. **Hold the 4 mm air gap tightly.** Check the spacer height on the built board: a 0.3 mm error in a 4 mm gap moves the driven-parasitic coupling noticeably.
5. **Re-tune the element as a true double-resonance stacked patch** in a unit-cell (periodic) simulation before touching the network:
    - Separate the two resonances: one near 4.5 GHz and one near 6.5 GHz. That needs the driven patch, the parasitic and the gap (3–6 mm) to be tuned together. Scaling one patch alone only moves a single narrow window (variants C–E).
    - Use the same parasitic size on every row unless a row-by-row taper is deliberately part of the design. The present 16.2–17.8 mm grading detunes the rows from one another.
    - Aim for a tight loop around the chart centre over 4–7 GHz, then use one quarter-wave transformer per element to bring it to 100 Ω.

6. **Change the feed type if 4–7 GHz at VSWR < 2 is required.** 54 % bandwidth is very hard for a direct edge-fed stacked patch. Proven routes for this bandwidth are:
    - aperture-coupled stacked patch (H-shaped or cross slots for dual polarisation), with the feed on a separate layer below the ground;
    - L-probe or meandered-probe feed into a thick air/foam patch;
    - a probe-fed stacked patch with capacitive-disc compensation.

   All of these also improve V/H isolation; the edge feeds here couple at only about −7 dB at 7 GHz.
7. **If you keep the edge feed:** add an inset or a λ/4 transformer from the edge resistance (100–300 Ω) to 100 Ω on each feed. That centres the existing narrow band on one frequency. It will not widen it to 4–7 GHz.
8. **Reduce the pitch** to ≤ 0.7 λ at 7 GHz (≈ 30 mm) if the upper band must be free of grating lobes. This needs the smaller element that a wideband feed allows.
9. **Measurement check:** measure S11 of one element on a test coupon and the bare network loaded with 100 Ω resistors. That separates element and network on the bench and confirms this analysis.

10. Limits of this analysis

- The element was simulated alone on a 60 mm ground. Mutual coupling in the array shifts the embedded impedance a little. The full-array model includes it.
- The full-array mesh (0.5 mm) snaps the 0.73 mm lines to 0.5 or 1.0 mm, which shifts the network's own resonances by a few percent.
- Copper thickness, solder mask, connector transitions and the spacers holding the 4 mm gap are not modelled.
- No measured data was available. The comparison is with the reported symptom (VSWR > 3, gain OK).

11. Reproduce

```
bash tools/install_openems.sh
/opt/openEMS/venv/bin/pip install ezdxf shapely matplotlib
cd projects/15_cband-6x6-dualpol-stacked-patch_match-debug/02_openems
source /opt/openEMS/venv/bin/activate
python sim_element.py --tag element_air                # as built (~5 min, 4 cores)
python sim_element.py --tag element_nopar --no-par
python sim_element.py --tag element_par1.25 --par-scale 1.25
python match_study.py
python sim_array.py --port 1 ; python sim_array.py --port 2   # full board, hours
python sim_wb_element.py --single --set "G=50,dp=10,Lp1=22.09,Lp2=20.77,h1=3.1,h2=4.45,Ls=16.1,Ws=0.92,Lst=5.79" --tag
wbN4
python opt_wb_element.py --gens 8                          # redesign optimiser (4 runs per generation)
python match_wb.py ; python match_wb.py --apply wbN4       # feed-layer matching on every candidate
python make_figures.py ; python plot_results.py ; python wb_figures.py wbN4 ; python build_report.py
```